Shubhankar Nehra

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Education

Virginia Tech — Blacksburg, VA
B.S. in Computer Science, Minor in Cybersecurity | GPA: 3.63 — *Expected May 2028*

Experience

Faith Innovations – Full-Stack Developer — Remote
Faith CRM – Medical Equipment Company (200 employees) — *May 2026 – Aug 2026*

- Architected and shipped a CRM (Next.js 15, TypeScript, Supabase/PostgreSQL) that replaced spreadsheets for a 20+ member sales team across India, saving each member 5.5 hours per week on average.
- Reached 100% user adoption within 60 days at the 200-employee company and gave leadership one live view of the $4.5M pipeline across 145 hospitals.
- Designed 2 configurable sales pipelines with cron-automated reminders and a map view of the 11 installations.
- Enforced least-privilege access at the database layer with Postgres Row-Level Security (RLS) policies for 3 roles.

Spacenet – Undergraduate Research Assistant — Blacksburg, VA
Backend Developer – LEO Satellite Network Testbed — *Feb 2026 – May 2026*

- Ported a Linux-only testbed simulating packet loss and routing across 1800+ moving low-Earth-orbit satellites to Windows for the first time, resolving cross-platform dependencies so every team member could onboard.
- Wrote 10+ pytest-mock unit tests stubbing a 2-3 hour Phase 1 run and an 18-hour Mininet simulation, cutting test runs to under 100 ms.
- Rebuilt the backend for local deployment of the compute-heavy simulation, removing unneeded multi-user JWT login and routing output to per-experiment workspaces.

DBWorkout – Undergraduate Research Assistant — Blacksburg, VA
Backend Developer – Secure Authorization — *Aug 2025 – Feb 2026*

- Hardened 60+ REST API endpoints as the only security-focused developer on a gamified SQL learning platform (Flask, PostgreSQL) whose backend serves 300+ concurrent students, TAs and faculty.
- Closed an information leak exposing raw PostgreSQL errors and table names by adding a global error handler that returns sanitized JSON with a traceable error ID.
- Replaced hard-coded role conditionals with reusable RBAC decorators, so new roles don't need route rewrites.
- Authored the team's security roadmap: API rate limiting, input validation, JWT and HttpOnly cookie hardening.

Projects & Achievements

OneVoice | *Python, PyTorch, CUDA (Team of 4, VTHacks)* | *GitHub* | *Devpost* | *Website* — *Sep 2026*

- Tap a face on camera and hear only that person live in earbuds, using a pretrained speech-separation model.
- Root-caused 3 GPU crashes (Triton, cuDNN, SIGFPE) to run the model live at 212-223 ms per audio window.
- Added fire-alarm passthrough to the live audio, backed by 31 tests and zero false triggers on real speech.

Cyber Apocalypse 2026 CTF | *HackTheBox (Team)* | *Writeups* — *Jul 2026*

- First CTF: team placed 571/6744 (top 9%) globally; personally solved 2 web exploitation challenges.
- Hijacked authenticated sessions by exploiting broken session management (writeup).
- Achieved stored XSS past a Content Security Policy via JSONP on an allowlisted Google APIs domain (writeup).

CancerScan | *Python, TensorFlow, OpenCV (Team)* | *GitHub* — *Jul 2025 – Aug 2025*

- Trained a ResNet-50 transfer-learning CNN (Roboflow) to detect cancer in brain MRI scans at 92% validation accuracy; a hospital in Greater Noida, India asked about a custom deployment.

Technical Skills

Languages: Python, C/C++, Assembly (RISC-V), SQL, Java, TypeScript
Security: Burp Suite, Nmap, Wireshark, Ghidra
Systems & Tools: Linux, Bash, Docker, Git, GDB, Google Cloud Platform
Frameworks & Databases: Flask, Next.js, Supabase/PostgreSQL
Certifications: CompTIA A+, ISC2 Certified in Cybersecurity (CC), HackTheBox CPTS (in progress)